Misr University of Science and Technology
Faculty of Engineering
Computer and Software Engineering Department

CSE 302 Lecture 1, Software Engineering

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CSE 302 SOFTWARE ENGINEERING 3(2-0-3):

The course covers software design and development topics beyond introductory programming, introducing additional software design and development methods, Alternative operating system and programming environment, Software design and development methods enabling students to undertake larger and more complex programs for computer and electrical engineering applications.

Prerequisite: CSE 308

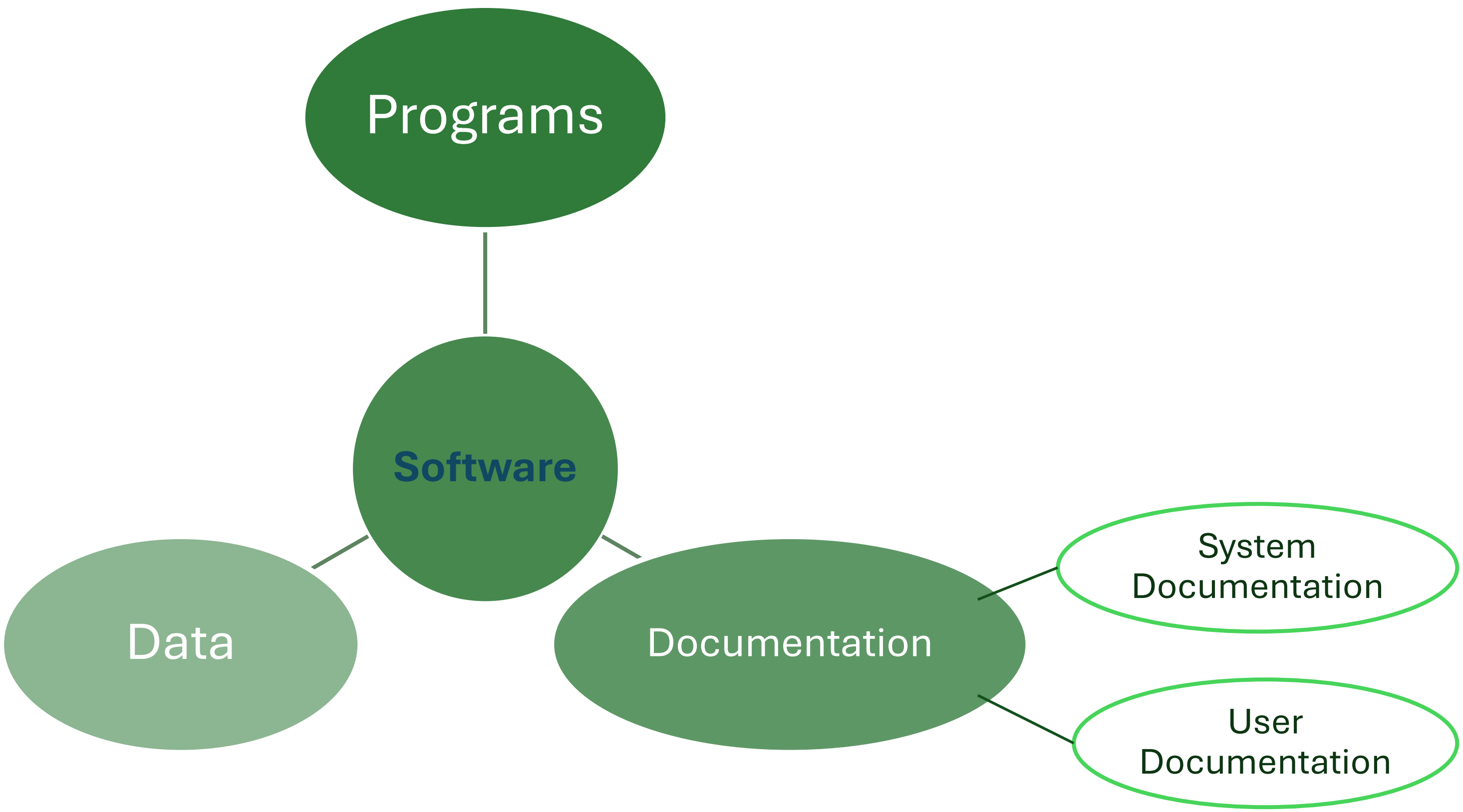
CSE 302	Software Engineering	3	2	-	3	5	CSE 308

Reference book

- The economies of ALL developed nations are dependent on software
- More and more systems are software controlled
- *Software engineering* is concerned with theories, methods and tools for professional software development
- Expenditure on software represents a significant fraction of GNP in all developed countries

Growth National Product

What is Software?



Software costs

- Software costs often dominate computer system costs. The costs of software on a PC are often greater than the hardware cost
- Software costs more to maintain than it does to develop. For systems with a long life, maintenance costs may be several times development costs
- Software engineering is concerned with cost-effective software development

Software products

✧ Generic products

- Stand-alone systems that are marketed and sold to any customer who wishes to buy them
- Examples – PC software such as graphics programs, project management tools; CAD software; software for specific markets such as appointments systems for dentists

✧ Customized products

- Software that is commissioned by a specific customer to meet their own needs
- Examples – embedded control systems, air traffic control software, traffic monitoring systems

Product specification

✧ Generic products

- The specification of what the software should do is owned by the software developer and decisions on software change are made by the developer

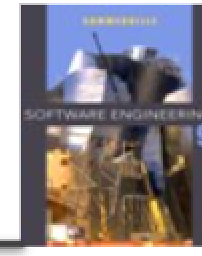
✧ Customized products

- The specification of what the software should do is owned by the customer for the software and they make decisions on software changes that are required

Frequently asked questions about software engineering

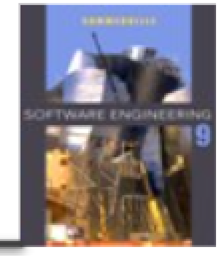
Question	Answer
What is software?	Computer programs and associated documentation. Software products may be developed for a particular customer or may be developed for a general market.
What are the attributes of good software?	Good software should deliver the required functionality and performance to the user and should be maintainable, dependable and usable.
What is software engineering?	Software engineering is an engineering discipline that is concerned with all aspects of software production.
What are the fundamental software engineering activities?	Software specification, software development, software validation and software evolution.
What is the difference between software engineering and computer science?	Computer science focuses on theory and fundamentals; software engineering is concerned with the practicalities of developing and delivering useful software.
What is the difference between software engineering and system engineering?	System engineering is concerned with all aspects of computer-based systems development including hardware, software and process engineering. Software engineering is part of this more general process.

Frequently asked questions about software engineering



Question	Answer
What are the key challenges facing software engineering?	Coping with increasing diversity, demands for reduced delivery times and developing trustworthy software.
What are the costs of software engineering?	Roughly 60% of software costs are development costs, 40% are testing costs. For custom software, evolution costs often exceed development costs.
What are the best software engineering techniques and methods?	While all software projects have to be professionally managed and developed, different techniques are appropriate for different types of system. For example, games should always be developed using a series of prototypes whereas safety critical control systems require a complete and analyzable specification to be developed. You can't, therefore, say that one method is better than another.
What differences has the web made to software engineering?	The web has led to the availability of software services and the possibility of developing highly distributed service-based systems. Web-based systems development has led to important advances in programming languages and software reuse.

Essential attributes of good software



Product characteristic	Description
Maintainability	Software should be written in such a way so that it can evolve to meet the changing needs of customers. This is a critical attribute because software change is an inevitable requirement of a changing business environment.
Dependability and Security	Software dependability includes a range of characteristics including reliability, security and safety. Dependable software should not cause physical or economic damage in the event of system failure. Malicious users should not be able to access or damage the system.
Efficiency	Software should not make wasteful use of system resources such as memory and processor cycles. Efficiency therefore includes responsiveness, processing time, memory utilisation, etc.
Acceptability	Software must be acceptable to the type of users for which it is designed. This means that it must be understandable, usable and compatible with other systems that they use.

Software engineering

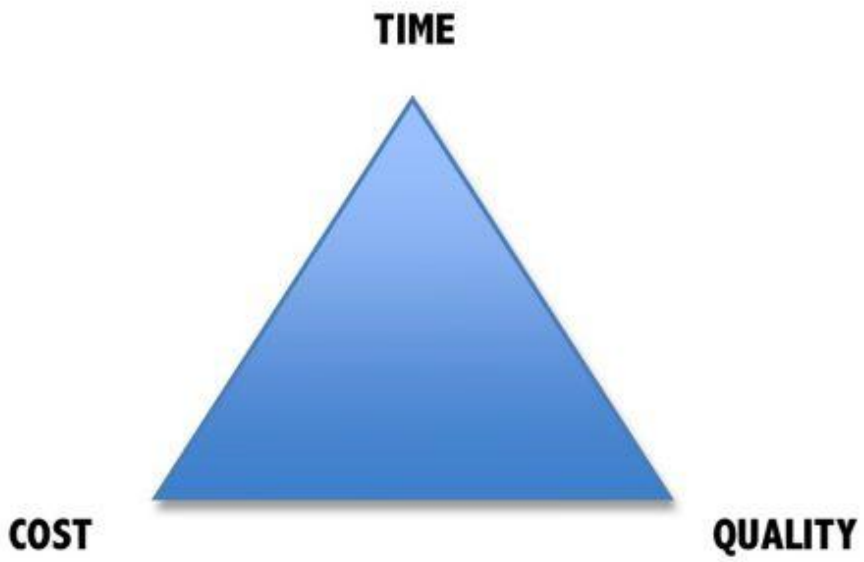
- ✧ Software engineering is an engineering discipline that is concerned with all aspects of software production from the early stages of system specification through to maintaining the system after it has gone into use.
- ✧ Engineering discipline
  - Using appropriate theories and methods to solve problems bearing in mind organizational and financial constraints.
- ✧ All aspects of software production
  - Not just technical process of development. Also project management and the development of tools, methods etc. to support software production.

Importance of software engineering

- ✧ More and more, individuals and society rely on advanced software systems. We need to be able to produce reliable and trustworthy systems economically and quickly.
- ✧ It is usually cheaper, in the long run, to use software engineering methods and techniques for software systems rather than just write the programs as if it was a personal programming project. For most types of system, the majority of costs are the costs of changing the software after it has gone into use.

Why is Software Engineering important?

Complex systems need a disciplined approach for designing, developing and managing them.



Software Development Crises

Projects were:

- Late.
- Over budget.
- Unreliable.
- Difficult to maintain.
- Performed poorly.

Software Crisis

Example 1: 2009,Computer glitch delays flights

Saturday 3rd October 2009-London, England (CNN)

- Dozens of flights from the UK were delayed Saturday after a glitch in an air traffic control system in Scotland, but the problem was fixed a few hours later.
- The agency said it reverted to backup equipment as engineering worked on the system.
- The problem did not create a safety issue but could cause delays in flights.
- Read more at: http://edition.cnn.com/2009/WORLD/europe/10/03/uk.flights.delayed

Software Crisis

Example 2: Ariane 5 Explosion

- European Space Agency spent 10 years and $7 billion to produce Ariane 5.
- Crash after 36.7 seconds.
- Caused by an overflow error. Trying to store a 64-bit number into a 16-bit space.
- Watch the video: http://www.youtube.com/watch?v=z-r9cYp3tTE

Software Crisis

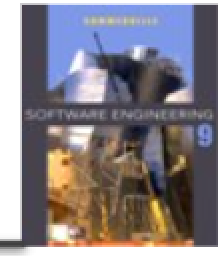
Example 3: 1992, London Ambulance Service

- Considered the largest ambulance service in the world.
- Overloaded problem.
- It was unable to keep track of the ambulances and their statuses. Sending multiple units to some locations and no units to other locations.
- Generates many exceptions messages.
- 46 deaths.

Software process activities

- Software specification, where customers and engineers define the software that is to be produced and the constraints on its operation
- Software development, where the software is designed and programmed
- Software validation, where the software is checked to ensure that it is what the customer requires
- Software evolution, where the software is modified to reflect changing customer and market requirements

General issues that affect most software



✧ Heterogeneity

- Increasingly, systems are required to operate as distributed systems across networks that include different types of computer and mobile devices

✧ Business and social change

- Business and society are changing incredibly quickly as emerging economies develop and new technologies become available. They need to be able to change their existing software and to rapidly develop new software

✧ Security and trust

- As software is intertwined with all aspects of our lives, it is essential that we can trust that software

Software engineering diversity

- There are many different types of software system and there is no universal set of software techniques that is applicable to all of these
- The software engineering methods and tools used depend on the type of application being developed, the requirements of the customer and the background of the development team

Application types

- ✧ Stand-alone applications
  - These are application systems that run on a local computer, such as a PC. They include all necessary functionality and do not need to be connected to a network.
- ✧ Interactive transaction-based applications
  - Applications that execute on a remote computer and are accessed by users from their own PCs or terminals. These include web applications such as e-commerce applications.
- ✧ Embedded control systems
  - These are software control systems that control and manage hardware devices. Numerically, there are probably more embedded systems than any other type of system.

Application types

- Batch processing systems
  - These are business systems that are designed to process data in large batches. They process large numbers of individual inputs to create corresponding outputs.
- Entertainment systems
  - These are systems that are primarily for personal use and which are intended to entertain the user
- Systems for modeling and simulation
  - These are systems that are developed by scientists and engineers to model physical processes or situations, which include many, separate, interacting objects

Application types

- Data collection systems
  - These are systems that collect data from their environment using a set of sensors and send that data to other systems for processing
- Systems of systems
  - These are systems that are composed of a number of other software systems

Thank you